Segurança em Redes de Comunicações

Simulation Environment Setup

&

Corporate Networks Fundamentals

Simulation Environment Setup

The environment setup can be deployed in Linux, Windows and MacOS (x86_64 and arm).
Linux users can build the simulation setup natively. Windows and MacOS users must build the simulation environment in a Linux virtual machine. The virtualization environment suggested is VirtualBox.
Any Linux (native or VM) can be used, however the assignments will focus on Debian based systems. A VirtualBox VM for Debian Linux (x86_64 and arm) is provided with all required packages already installed.
The minimum recommend for the VM is 8GB of RAM and two CPU cores, if possible use 16GB of RAM and four or more CPU cores. **Do not allocate more than 50% of the cores or RAM of the host PC to the VM.**

All future references to PC operations refer to the Linux setup (native or VM)

Traffic Monitoring (with Wireshark)

1. Open Wireshark, root permissions should not be required (see user group above). Start a capture on the interface that provides Internet to your PC. Open your browser and access your favorite sites. Stop the capture and analyze the packets.
>> Try to identify the packets exchanged between your PC and the servers.
>> Try to identify used protocols and how are they encapsulated at different layers.
>> Identify the IP addresses of the hosts that exchanged packets.
You may save the capture with File→Save and reopening a previous capture with File→Open.

2. Start a capture on the interface that provides Internet to your PC. Perform a set of connectivity tests (pings). Stop the capture and analyze the packets.
>> Try to identify the packets exchanged.
>> Try to identify used protocols and how are they encapsulated at different layers.
>> Identify the IP addresses of the hosts that exchanged packets.

3. Start a capture on the interface that provides Internet to your PC. Go to YouYube and play a video. Stop the capture and analyze the packets. If possible, repeat the capture using a different browser (Firefox or Chrome/Chromium).
>> Try to identify the packets exchanged.
>> Try to identify used protocols and how are they encapsulated at different layers.
>> Identify the IP addresses of the hosts that exchanged packets.

4. Open the previous captures and apply the following visualization filters:
- IPv4 packets sent or received by your PC: ip.addr==<ipv4_address>
- IPv4 packets sent by your PC: ip.src==<ipv4_address>
- IPv4 packets received by your PC: ip.dst==<ipv4_address>
- Only ICMP packets: icmp
- ICMP **and** ARP packets: icmp **or** arp
- Only TCP packets: tcp
- Only HTTPS (TCP port 443): tcp.port==443

Note: Replace <ipv4_address> by your PC IPv4 address.
Note2: Wireshark has capture and visualization filters. At this stage do do not use capture filters.

5. Open one of the previous captures and explore the traffic analysis features from Wireshark. From Wireshark menu:
- Statistics → Endpoints
- Statistics → Conversations
- Statistics → I/O Graphs

Network Deployment Fundamentals

1. Open GNS3. At (Preferences-General), verify/setup all storing and program paths, avoiding paths with spaces and non ASCII characters.

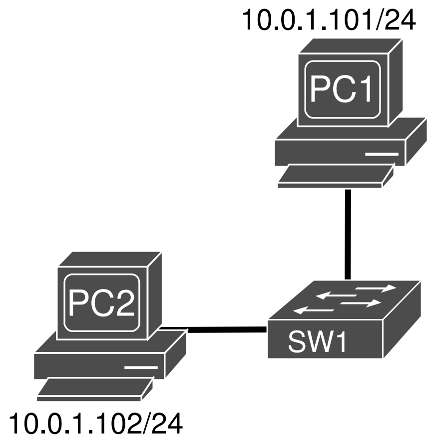
2. Fist step is to create a simple Linux terminal inside GNS3. The best option is to use a Linux busybox docker.
If your VM/GNS3 does not have a Docker busybox template, do:
In GNS3, go to Edit → Preferences → Docker containers → New.
Choose “new image”, and define “image name” as **busybox**, press next until the end.
Add the directory /etc/network/ as permanent in the advanced tab of the GNS3 template configuration. After, the file /etc/network/interfaces can be use to make the network configuration permanent. Example:

```
auto eth0
iface eth0 inet static
        address 10.0.1.101
        netmask 255.255.255.0
        gateway 10.0.1.1
```

Warning: If a GNS3 project fails to open with an error relate with a Docker container, delete all unused docker containers with:

```
$ docker container prune
```

3 Create a new Blank Project (File menu or CTRL+N) and give it a name. Add Two Virtual PC (busybox) and one basic Layer 2 Switch (Ethernet Switch) to your project. Rename the PCs (PC1 and PC2), right-clink and option “Change hostname”.
Note: A basic Layer 2 Switch (Ethernet Switch) does not run Spanning-Tree Protocol and should not be used in redundant Layer 2 networks (with connection loops).



4. Interconnect them and configure their IPv4 address according to the above network diagram. Start all nodes in the project, double left-click on the PCs to open their consoles, and verify the network interfaces and respective MAC addresses:

```
PC# ip addr
```

and configure their IPv4 addresses:

```
PC1# ip addr add 10.0.1.101/24 dev eth0
...
PC2# ip addr add 10.0.1.102/24 dev eth0
```

Verify the final busybox IPv4 configuration with the command:

```
PC# ip addr
```

5. Start a capture on the link between PC1 and SW1 (Right-click-Start Capture). From PC1 console test connectivity with PC2 with the ping command (and vice-versa):

```
PC1# ping 10.0.1.102
...
PC2# ping 10.0.1.101
```

>> Analyze the captured packets. Namely, Ethernet header (MAC address) and IPv4 header (IPv4 addresses).

If not using the provided VM:

Download the following Cisco routers' firmware: (i) Router 7200 Firmware 15.1(4) IOS Image, and (ii) Router 3725 Firmware 12.4(21) IOS Image.

At (Preferences-Dynamips-IOS Routers") create two new router templates ("New" button on the bottom left):

- **Router 7200** - recommended IOS image: 7200 with IOS 15.1(4) and network adapters C7200-IO- 2FE and PA-2FE-TX (4 FastEthernet → F0/0,F0/1+F1/0,F1/1), all other values can be the default ones;
- **Switch L3** – will be a router 3725 with IOS image 12.4(21) and adapters GT96100-FE and NM-16ESW (1 FastEthernet + 16 port switch module). Choose option "This is an EtherSwitch router" when defining the device platform, all other values can be the default ones.

6. The definition of the "Idle-PC" value will allow the host machine to assign the correct amount of resources to the virtual devices. <u>You must repeat this procedures every time your PC CPU reaches values higher than 90%.</u> Check the CPU utilization top command in Linux.

To define the "Idle-PC" value:

- Click "Idle-PC finder" during template setup, OR
- Add router to project, start it (should be the only one ON), open console (wait for prompt), left click the device and choose option "Auto Idle-PC", OR
- Add router to project, start it (should be the only one ON), open console (wait for prompt), left click the device and choose option "Idle-PC", choose one value (prefer the ones marked with *) and verify the CPU utilization. If any "dynamips" process is using more than 5%-10% CPU choose another value.

This must be done for each router template, NOT each router! Each template will have a different "Idle-PC" value. All routers from the same template will share the same value.

Note 1: All devices from the same template must be equal in terms of virtual hardware.

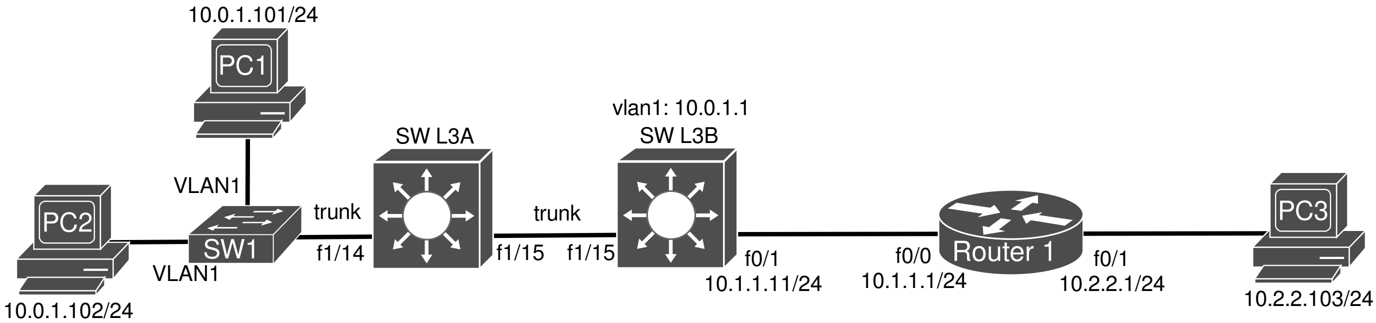
Note 2: After changing any device hardware characteristic or adding/removing network modules, the "Idle-PC" value must be changed in the template. If necessary, create a new template with different characteristics/modules.

At this phase your GNS3 installation should have (at least):

- 1 router Cisco c7200;
- An "EtherSwitch" (Layer 3 switch) based on a router c3725 with a 16 port switch module.

Layer 3 Switch and Router configurations

7. Add two Layer3 Switches (EtherSwitch routers), one router (c7200) and a third PC, and interconnect them according to the diagram below.



8. Place the cursor over Layer3 Switches (A and B) and identify (on the pop-up) the 16 ports numbered from FastEthernet x/0 to FastEthernet x/15, those are the ones of the switch module that must be used to connect to other switches using Trunk links (usually is F1/0 to F1/15).
Connect SW1 to port FastEthernet x/14, and use ports FastEthernet x/15 to interconnect SWL3A and B. The connection between SWL3B and Router1 is a Layer3 connection and must be done using Layer3 interfaces (non-switch port, e.g. FastEthernet 0/1).
Start both Layer3 switches and Router 1 by double right clinking it and choosing Start. Wait a few seconds, open the console of the switches by double left clinking it, and press enter to obtain prompt.
>> Use the command `show ip interface brief` to verify the status of the connected ports. The status and protocol (of the ports in use) should be up and up. If not, stop and start the equipment.
Note: a Layer3 switch Layer2 port (switch port) can be converted to a Layer3 (non-switch) port with the command `no switchport`.

9. To configure SWL3A (working has a Layer2 switch), verify the existence of VLAN1 on the switch database:

```
SWL3A# show vlan-switch
```

Second, configure ports f1/14 and f1/15 as trunks

```
SWL3A# configure terminal
SWL3A(config)# interface FastEthernet 1/15
SWL3A(config-if)# switchport mode trunk
SWL3A(config)# interface FastEthernet 1/14
SWL3A(config-if)# switchport mode trunk
SWL3A(config-if)# end
SWL3A# write
```

Note: To verify Cisco routers running configuration type: `show run`.

10. To configure SWL3B, it is necessary to configure the trunk port to SWL3A, configure the virtual interface of VLAN1 and Layer 3 interface F0/1 with IPv4 addresses, and activate dynamic IPv4 routing (OSPFv2 protocol).
Verify the existence of VLAN1 on the switch database:

```
SWL3B# show vlan-switch
```

Configure port F1/15 as trunks:

```
SWL3B# configure terminal
SWL3B(config)# interface FastEthernet 1/15
SWL3B(config-if)# switchport mode trunk
```

Activate IPv4 routing, and configure virtual interface VLAN1 address and routing:

```
SWL3B(config)# ip routing
SWL3B(config)# interface vlan 1
SWL3B(config-if)# no autostate
SWL3B(config-if)# ip address 10.0.1.1 255.255.255.0
SWL3B(config-if)# ip ospf 1 area 0
SWL3B(config-if)# no shut
```

Configure Layer3 interface with address and routing:

```
SWL3B(config)# interface f0/1
SWL3B(config-if)# ip address 10.1.1.11 255.255.255.0
SWL3B(config-if)# ip ospf 1 area 0
SWL3B(config-if)# no shut
SWL3B(config-if)# end
SWL3B# write
```

Note: To verify Cisco routers running configuration type: show run.

11. To configure Router1 (e.g., IP address/mask, activation of interfaces, and dynamic routing protocol - OSPFv2):

```
Router1# configure terminal
Router1(config)# interface FastEthernet 0/0
Router1(config-if)# ip add 10.1.1.1 255.255.255.0
Router1(config-if)# ip ospf 1 area 0
Router1(config-if)# no shutdown
Router1(config-if)# interface FastEthernet 0/1
Router1(config-if)# ip add 10.2.2.1 255.255.255.0
Router1(config-if)# ip ospf 1 area 0
Router1(config-if)# no shutdown
Router1(config-if)# end
Router1# write
```

Verify the equipment (Router1 and SWL3B) IPv4 routing tables with the command:

```
*# show ip route
```

>> Analyze IPv4 routing tables of SWL3B and Router1.
>> Test connectivity between all devices.

Note: You may save the router configuration in an external file; right-click the router and save configuration. Analyze the configuration (*.cfg) file created in your project folder.

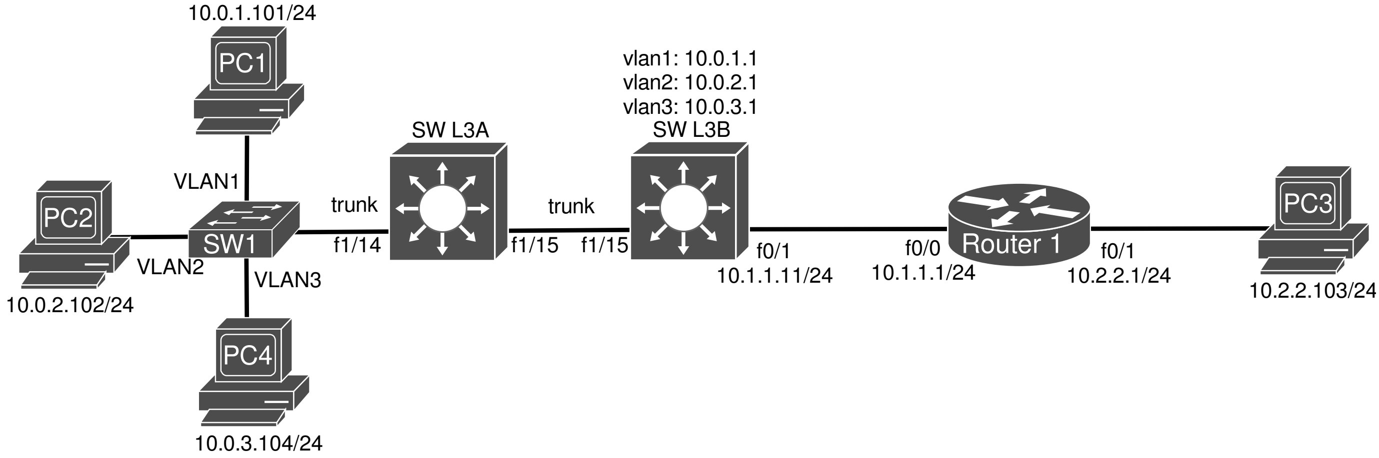
12. To configure SW1, double left-click and:
- Define the switch ports connected to PC1 and PC2 as VLAN1 and access ports, and
- Define the port connected to SW L3 A as trunk port (protocol 802.1Q - type dot1q).

13. To Configure the PCs IPv4 addresses and respective gateway.

```
PC1# ip addr add 10.0.1.101/24 dev eth0
PC1# ip route add default via 10.0.1.1
...
PC2# ip addr add 10.0.1.102/24 dev eth0
PC2# ip route add default via 10.0.1.1
...
PC3# ip addr add 10.2.2.103/24 dev eth0
PC3# ip route add default via 10.2.2.1
```

Use the command ip route to view the PCs' default gateways/ routing table.
Start a capture on the link between SWL3B and Router1 (Right-click-Start Capture). Test connectivity between all devices.
>> Analyze the captured packets. Namely, the OSPF and ICMP packets.
Note: A PC gateway is the IP address of the router in than LAN that interconnects it to other LAN.

VLAN Segmentation/Isolation



14. Reconfigure **both** SWL3A and SWL3B by adding VLAN2 and VLAN3 to their databases:

```
SWL3*# vlan database
SWL3*(vlan)# vlan 2
SWL3*(vlan)# vlan 3
SWL3*(vlan)# apply
SWL3*(vlan)# exit
SWL3*# show vlan-switch
```

Configure SWL3 B virtual interfaces for VLAN2 and VLAN3 (address and routing):

```
SWL3B(config)# interface vlan 2
SWL3B(config-if)# no autostate
SWL3B(config-if)# ip address 10.0.2.1 255.255.255.0
SWL3B(config-if)# ip ospf 1 area 0
SWL3B(config-if)# no shut
SWL3B(config-if)# interface vlan 3
SWL3B(config-if)# no autostate
SWL3B(config-if)# ip address 10.0.3.1 255.255.255.0
SWL3B(config-if)#ip ospf 1 area 0
SWL3B(config-if)#no shut
SWL3B(config-if)#end
SWL3A# write
```

Note: To verify Cisco routers running configuration type: show run.

>> Analyze IPv4 routing tables of SWL3B and Router1.

>> Test connectivity between all devices.

15. Add a fourth PC (PC4) to VLAN3, and change PC2 to VLAN2.

Note: A device VLAN depends on the VLAN configured on the access switch (and IPv4 address).

Reconfigure SW1, double left-click and (i) define the switch ports connected to PC2 as VLAN2 and access, and (ii) define the switch ports connected to PC4 as VLAN3 and access. (Re)Configure PC2 and PC4 IPv4 addresses (without gateway configured):

```
PC2# ip addr delete 10.0.1.102/24 dev eth0
PC2# ip addr add 10.0.2.102/24 dev eth0
...
PC4# ip addr add 10.0.3.104/24 dev eth0
```

Test connectivity between PC2 and PC4. Start a packet capture on two PC links from different VLANs, and ping between those devices. Analyze the (non) captured packets, namely the ARP packets.

>> Explain the purpose of Layer 2 VLAN segmentation.

16. Configure PC1, PC2 and PC4 IPv4 gateways:

```
PC1# ip route add default via 10.0.1.1
...
PC2# ip route add default via 10.0.2.1
...
PC4# ip route add default via 10.0.3.1
```

Test connectivity between the terminals.

>> Explain how the VLAN segmentation (at Layer3/IP level) is now broken.

>> In which network devices can now a Layer 3 VLAN segmentation be implemented?

17. Start a capture on the link between SWL3A and B, from PC2 and PC4 ping their gateways.

>> Analyze the captured packets. Namely, the ICMP packets' Ethernet header (802.1Q protocol).

>> How Layer2 packets from different VLAN are discriminated in a trunk link?

18. Trunk links can be restricted to some VLAN traffic. On the trunk link between SWL3A and B allow only traffic from VLAN1 and VLAN2:

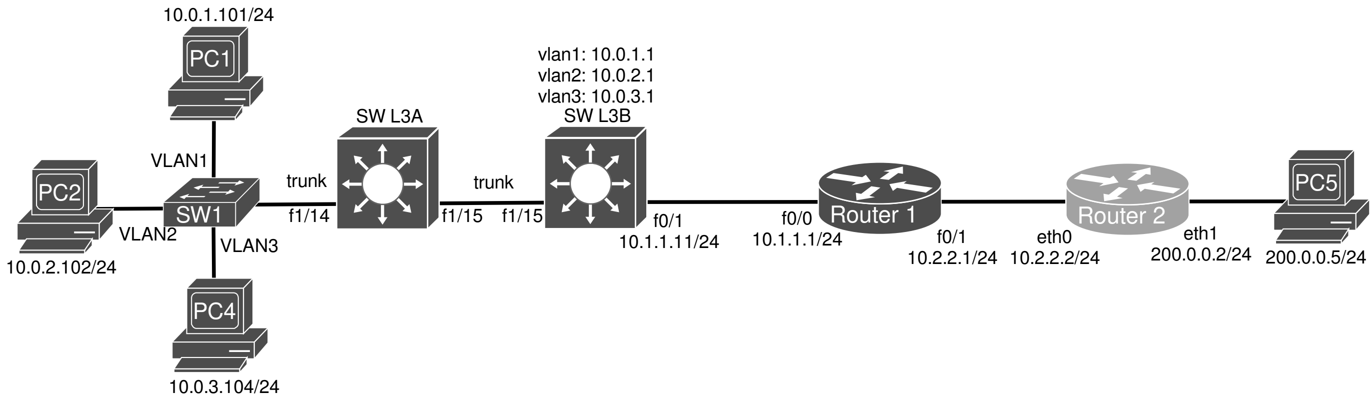
```
SWL3*# configure terminal
SWL3*(config)# interface FastEthernet 1/15
SWL3*(config-if)# switchport trunk allowed vlan 1,2,1002-1005
```

Note: VLAN 1002-1005 permissions are required by Cisco mechanisms and cannot be be restricted.

>> Test connectivity between all devices. Verify if PC4 has connectivity with its gateway. Verify if PC4 has connectivity with the other VLANs devices.

>> Explain the purpose of the trunk permissions when implementing VLAN segmentation and/or VLAN isolation.

Network Border Router/Firewall (NAT/PAT)



19. A Linux Router/Firewall may be implemented as a **docker** Linux container with the [FRRouting protocol suite] installed. With the provided Dockerfile, create a custom docker image:

```
$ docker build -t local/frrouting -f Dockerfile_frrouting .
```

Add the Docker container to GNS3 as template:
- Choosing an existing image;
- Define it as local/frrouting container image;
- Activate 4 network interfaces (adapters), and
- **Add the following directories as permanent in the advanced tab:**
  /etc/frr/
  /etc/conntrackd
  /etc/init.d/

Note: FRRouting image is a linux container with a set of services/protocols installed that enable advance network features. The command `vtysh` provides a Cisco-like CLI.

Note2: This Linux bash commands are not permanent! They will disappear upon a reboot. To make them permanent write them in file **/etc/init.d/adv_network**. These commands will be run on every start of the Docker container.

Note3: Running this docker outside GNS3 requires additional privileges enabled (NET_ADMIN and SYS_ADMIN): `$ docker run --cap-add NET_ADMIN --cap-add SYS_ADMIN -it local/frrouting`

20. Add a Linux router (Router2) as the network Internet access network device using the FFRouting Docker device.
Remove PC3 and replace it PC5 (simulating an Internet device, in a public IPv4 network).
To configure PC5:

```
PC5# ip addr add 200.0.0.5/24 dev eth0
PC5# ip route add default via 200.0.0.2
```

21. To configure Router2:

```
bash-5.1# vtysh
Router2# configure terminal
Router2(config)# int eth1
Router2(config-if)# ip address 200.0.0.2/24
Router2(config-if)# no shutdow
Router2(config)# int eth0
Router2(config-if)# ip address 10.2.2.2/24
Router2(config-if)# ip ospf 1 area 0
Router2(config-if)# no shutdow
Router2(config-if)# router ospf 1
Router2(config-if)# end
Router2# write
```

Verify the IPv4 routing table at Router2 using the commands:

```
PE1#show ip route
PE1#show ip fib
```

>> Test connectivity between the PCs.
>> Verify the routing table on SWL3B and Router1 IPv4 Routing table: show ip route
>> Explain the lack of connectivity to PC5 (Internet).
Note: The IPv4 routing between the network and the Internet can not be the same internal routing process. May be another OSPF process, another protocol, or static routing.

22. Router 2 must dynamically announce a default route for all internal routers:

```
bash-5.1# vtysh
Router2# configure terminal
Router2(config)# router ospf 1
Router2(config-router)# default-information originate always
Router2(config-if)# end
Router2# write
```

>> Verify the new entry on SWL3B and Router1 IPv4 Routing table: show ip route
>> Capture packets in the link between Router2 and the internet device (PC5).
>> Test connectivity between the PC1/2/4 and PC5.
>> Verify that the packets source/destination addresses in the Internet.

23. **Packets with a IPv4 private address as source must never flow on public networks (Internet).** It is required to activate address and port translation mechanisms (NAT/PAT) in Router 2. Assume that the network will use the IPv4 public address 192.0.0.100 to 192.0.0.110. In bash CLI:

```
# iptables -t nat -A POSTROUTING -j SNAT -o eth1 -s 10.0.0.0/8 --to 192.0.0.100-192.0.0.110
```

Verify the applied rules:

```
# iptables -t nat -L -vn
```

Ping PC5 from PC1/2/4 and verify the correct translation of the source IPv4 addresses. Use the following command to verify the active NAT/PAT translations:

```
# conntrack -L -n
```

>> What can you conclude from the NAT/PAT translations table?
>> Capture packets in the link between Router2 and the internet device (PC5). Verify that the packets source/destination addresses have been changed.
Note: the network 192.0.0.0/24 must be announced to the Internet. Otherwise, no route exists to the virtual NAT/PAT network.

24. Save iptables rules using:

```
# iptables-save > /etc/init.d/iptables.rules
```

To restore iptables rules use:

```
# iptables-restore < /etc/init.d/iptables.rules
```

Note: The last command may be added to **/etc/init.d/adv_network** to run on boot.

Appendix – Create a Debian VM

If using other Debian based Linux VM (already with Desktop installed), follow the following steps to create the base environment setup.
List of Linux required packages:
- vim, nano, geany, curl, wget;
- wiresahrk, tshark;
- docker.io, docker-compose.

Install commands:

```
sudo apt -y install curl wget geany vim nano
sudo apt -y install wireshark tshark
sudo apt -y install docker.io docker-compose
```

The local user must be added to groups wireshark and docker:

```
sudo usermod -aG wireshark $USER
sudo usermod -aG docker $USER
```

A logout/login is required to update group members.

GNS3 packages:
- gns3-gui, gns3-server, python3-pyqt6*;
- ca-certificates gnupg2 busybox-static;
- dynampis, ubridge.

Install commands:

```
sudo apt -y install python3 python3-pip pipx python3-pyqt6 python3-pyqt6.qtwebsockets python3-pyqt6.qtsvg
sudo apt -y install ca-certificates gnupg2 busybox-static
pipx install gns3-server
pipx install gns3-gui
pipx inject gns3-gui gns3-server PyQt6
pipx ensurepath
source ~/.bashrc
```

Download dynampis and ubridge deb packages from Ubuntu GNS PPA:
For x86_64: ubridge and dynamips
For arm64: ubridge and dynamips
Then run:

```
sudo dpkg -i dynamips*.deb
sudo dpkg -i ubridge*.deb
sudo usermod -aG ubridge labredes
rm dynamips*.deb
rm ubridge*.deb
```